

CASTEAU, Belgium

WMO: 064325

Lat: 50.503N

Lon: 3.983E

Elev: 256

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.4	25.1	13.8	11.2	21.8	19.5	14.8	29.1	18.8	49.3	16.8	48.8	4.9	40	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	86.1	67.9	82.2	66.0	78.7	64.2	69.3	81.8	67.5	78.9	65.7	74.9	5.8	170

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	92.8	72.4	64.1	90.5	71.5	62.5	85.3	69.7	33.5	83.9	32.0	79.1	30.7	74.9	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
15.2	13.3	11.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	16.4	73.2	5.3	1.7	12.6	74.4	9.5	75.3	6.5	76.3	2.6	77.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	38.8	39.3	45.1	51.0	56.0	62.7	66.2	64.9	59.4	53.8	45.7	41.1
	DBStd	11.49	6.82	7.29	6.18	7.00	6.62	5.83	5.69	5.38	5.14	6.40	6.67	6.93
	HDD50	1383	347	302	171	70	14	0	0	0	0	34	162	283
	HDD65	4949	811	721	616	423	290	113	54	64	186	350	580	741
	CDD50	2140	1	1	21	99	200	382	502	463	281	151	32	7
	CDD65	230	0	0	0	2	11	45	92	62	16	2	0	0
	CDH74	1999	0	0	0	31	107	424	771	521	135	10	0	0
	CDH80	560	0	0	0	0	20	113	221	173	33	0	0	0
Wind	WSAvg	5.1	6.0	6.1	5.6	5.0	4.7	4.4	4.4	4.4	4.2	4.7	5.2	6.1
Precipitation	PrecAvg	31.9	2.8	2.4	2.1	1.9	2.4	3.0	3.0	3.3	2.4	2.7	3.0	3.1
	PrecMax	39.8	6.4	5.4	5.4	4.7	4.6	6.1	5.9	7.5	6.7	5.2	5.0	6.6
	PrecMin	24.8	0.2	0.6	0.1	0.0	0.4	1.1	0.9	0.6	0.6	0.4	0.4	0.8
	PrecStd	3.9	1.4	1.1	1.2	1.1	1.1	1.2	1.4	2.0	1.4	1.1	1.1	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	60.4	67.8	78.4	83.9	89.4	91.4	91.5	84.4	75.2	64.5	57.3
		MCWB	50.9	48.4	52.4	60.0	64.3	70.0	69.0	69.3	65.7	62.4	57.8	52.3
	2%	DB	53.2	53.7	64.0	73.5	77.1	82.7	85.9	84.4	77.4	70.0	60.8	55.3
		MCWB	50.1	47.9	51.9	56.4	61.3	66.7	67.1	67.6	63.7	61.3	56.5	52.1
	5%	DB	50.2	50.4	59.1	68.2	73.2	78.8	82.0	79.1	71.9	66.0	57.6	53.3
		MCWB	47.6	47.2	49.7	55.6	59.5	64.6	65.7	65.4	62.4	58.7	53.7	50.0
	10%	DB	48.2	48.6	55.1	62.7	68.3	75.0	78.5	75.2	68.2	62.7	55.3	50.4
		MCWB	46.2	46.0	48.3	53.0	58.1	63.3	64.5	63.5	59.3	57.7	52.1	47.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	50.7	54.6	60.8	67.5	71.0	70.9	72.6	68.3	64.0	60.3	54.4
		MCDB	54.2	53.8	64.4	74.4	78.8	82.7	83.6	85.4	80.8	72.6	63.5	56.7
	2%	WB	50.0	49.3	52.2	57.6	64.0	68.3	69.0	68.7	65.7	61.6	56.8	52.4
		MCDB	51.6	51.6	60.8	69.5	71.7	79.9	82.0	81.1	75.1	68.6	60.2	55.1
	5%	WB	47.9	47.8	50.7	55.5	61.2	65.9	67.4	66.8	63.4	59.7	54.5	50.6
		MCDB	49.8	50.5	57.2	66.1	70.3	75.6	78.7	76.8	70.7	64.8	57.7	53.0
	10%	WB	45.9	46.0	49.2	53.9	59.1	64.0	65.5	64.9	61.1	58.2	52.3	48.3
		MCDB	47.6	48.9	54.3	62.9	68.1	72.3	74.8	73.1	67.2	62.3	55.4	50.7
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.8	13.8	16.1	16.3	16.9	16.5	16.4	15.2	12.2	9.3	8.1
		MCDBR	8.9	12.5	22.4	23.3	22.5	23.7	22.9	23.2	21.1	16.5	11.0	8.2
		MCWBR	8.0	8.9	12.1	11.0	10.4	10.4	8.9	10.0	10.4	9.0	8.0	6.4
	5% WB	MCDBR	9.2	11.2	18.4	21.0	19.2	20.7	19.8	20.1	18.4	14.6	9.3	7.9
MCWBR		8.7	8.7	11.2	10.6	9.9	10.0	8.9	10.2	9.8	8.4	7.6	6.9	
Clear-Sky Solar Irradiance	taub	0.326	0.347	0.382	0.416	0.427	0.424	0.425	0.416	0.388	0.364	0.338	0.323	
	taud	2.395	2.357	2.258	2.180	2.194	2.226	2.225	2.272	2.337	2.402	2.395	2.366	
	Ebn at Noon	222	247	257	261	264	265	262	259	253	238	214	203	
	Edh at Noon	20	27	35	42	43	43	42	38	32	25	20	18	
All-Sky Solar Radiation	RadAvg	255	471	841	1334	1566	1715	1630	1385	1054	621	301	202	
	RadStd	31	92	115	151	150	158	180	140	95	53	35	30	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page